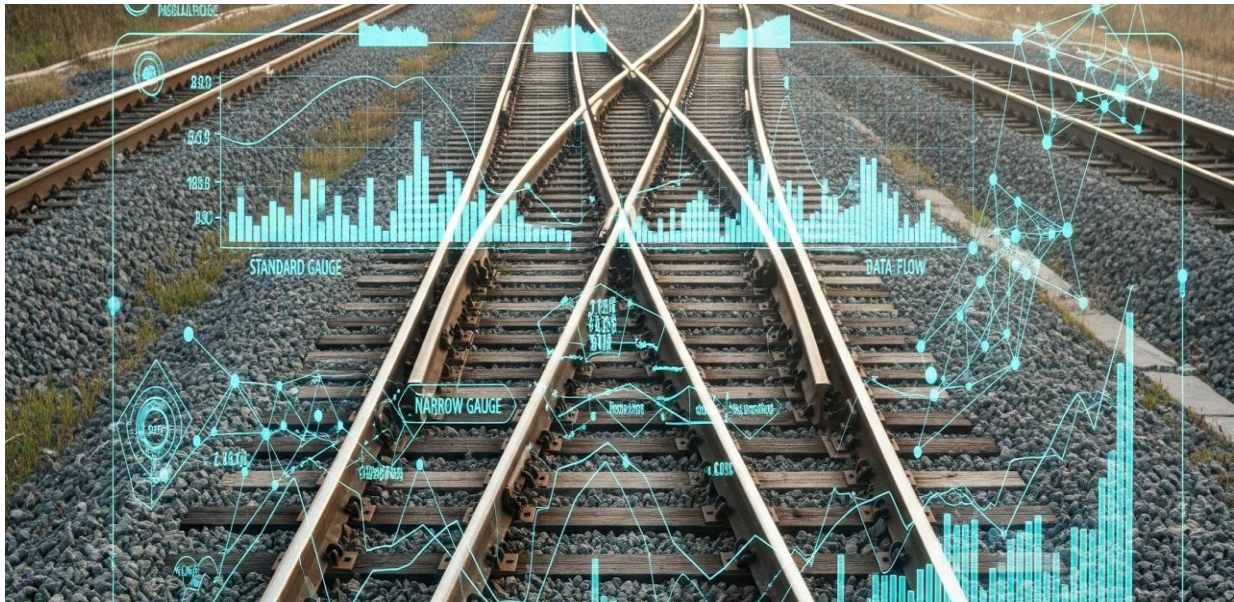




Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges.csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divi **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:

In this scenario, Load the CSV file into a Pandas DataFrame using **pandas** for easy data handling. Display the first five rows with `head()` and inspect column names to understand the dataset structure. Check for missing values using `isnull()` and replace them with 0 to ensure consistency. Finally, convert the gauge-related columns (Broad, Metre, Narrow, Total) into numeric types using `to_numeric()` so they can be used for analysis.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:

The dataset was successfully loaded and cleaned using **pandas**. Missing values were handled by replacing them with 0, ensuring no gaps in the data. Converting gauge columns to numeric types made the dataset consistent and ready for further analysis. Overall, the data is now structured, clean, and suitable for reliable insights.

Scenario 2: Simple Visualization (□ Easy)

Description:

In this Scenario, Use matplotlib to create a simple line graph by plotting Year against Total railway tracks. Add a title and axis labels to make the chart clear and informative. This visualization helps quickly observe overall growth and identify whether the trend is increasing or decreasing over time.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph :

The line graph plotted shows the change in total railway tracks over the years. The X-axis represents the years, while the Y-axis represents the total track length. If the line moves upward, it indicates an increasing trend in railway track growth; if it slopes downward, it shows a decrease. Overall, the graph provides a clear visual understanding of how railway infrastructure has developed over time.

Output:

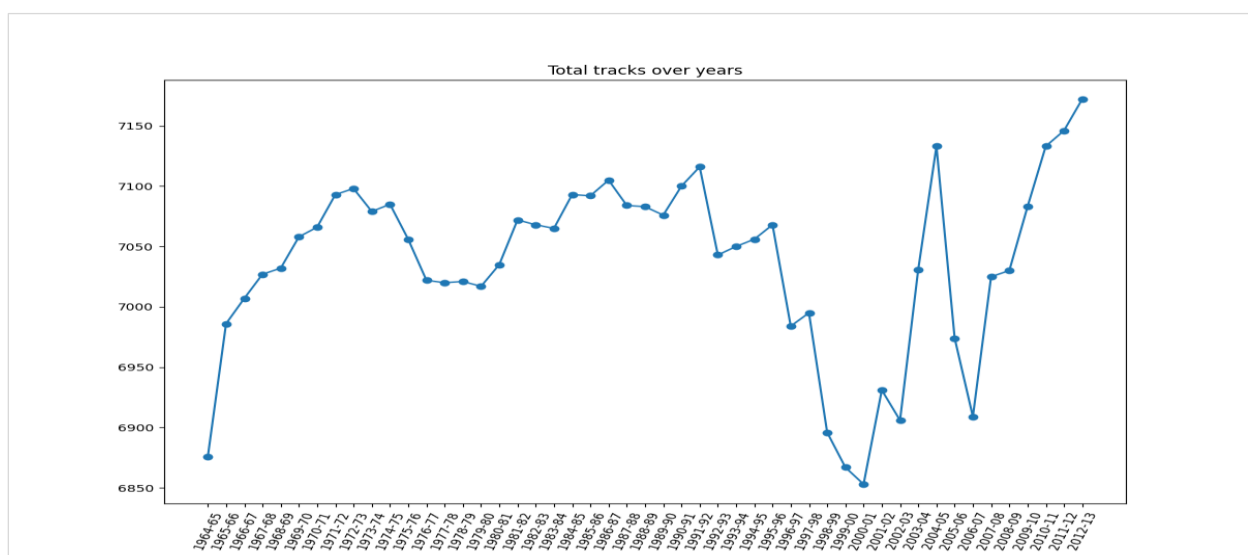


Fig 1: Total Track over Years

Conclusion:

The line graph shows the overall trend of railway track growth over the years. The Total track length generally indicates whether the network is expanding or declining. An upward trend suggests continuous development, while a downward trend indicates reduction. Overall, the visualization provides a quick and clear understanding of railway infrastructure changes.

Scenario 3: Filtering + Bar Chart (□ Medium)

Description:

In this Scenario, Filter the dataset for years after 2000 and select Broad, Metre, and Narrow gauge values for analysis. Using matplotlib, a grouped bar chart is created to compare all three gauges.

The bar chart includes proper labels and a legend for clarity. This helps visualize modern railway expansion and compare gauge usage effectively.

Task	Description
1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph :

The grouped bar chart created using matplotlib compares Broad, Metre, and Narrow gauge values for years after 2000. Each year shows three bars representing the different gauges, making comparison easy. The X-axis shows years and the Y-axis shows track values. From the chart, Broad Gauge typically appears highest, showing it dominates recent railway expansion.

Output :

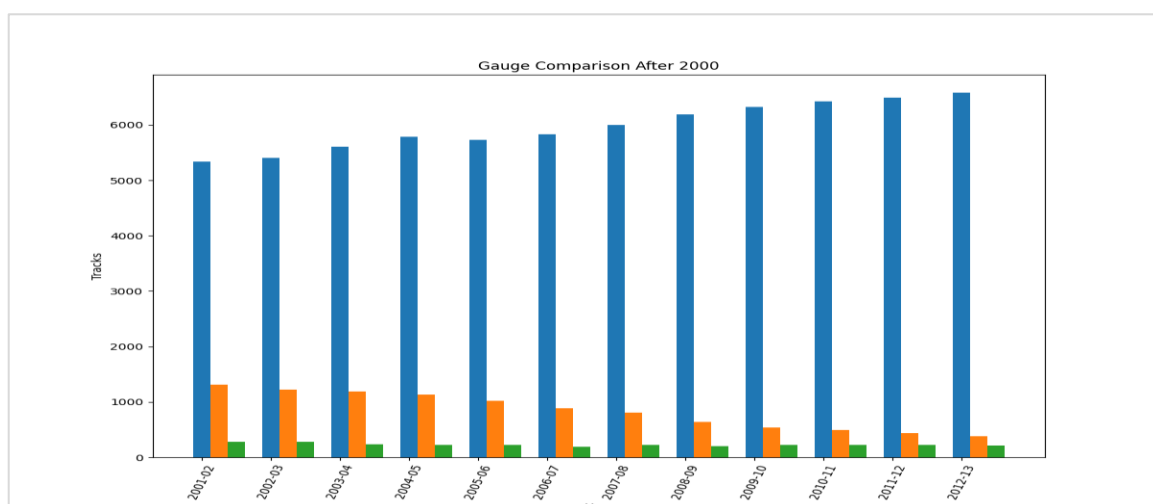


Fig 2: Gauge Comparison After 2000

Conclusion:

The analysis shows a clear preference for Broad Gauge in recent railway development. Using matplotlib, the grouped bar chart highlights that Broad Gauge consistently leads over Metre and Narrow gauges. This indicates standardization and modernization of the railway network. Overall, newer expansions are strongly dominated by Broad Gauge systems.

Scenario 4: Feature Engineering + Pie Chart (□ Medium)

Description:

In this Scenario, Compute the total values of Broad, Metre, and Narrow gauges across all years and organize them using pandas. Use matplotlib to plot a pie chart with percentage labels (autopct) to show each gauge's share. This visualization clearly represents the contribution of each gauge type and helps identify which one dominates overall usage.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph :

The pie chart generated using matplotlib shows the percentage share of Broad, Metre, and Narrow gauges. Each slice represents the contribution of a gauge type to the total railway network. Broad Gauge usually forms the largest portion, indicating its dominance in usage.

Output :

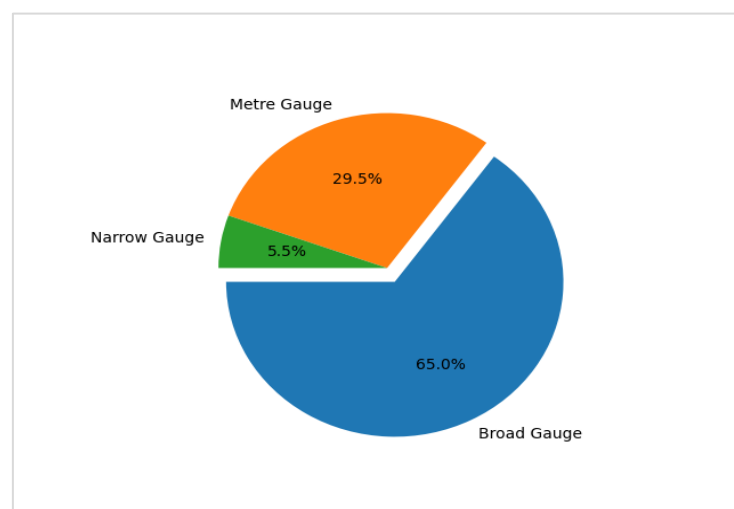


Fig 3: Percentage Contribution

Conclusion:

This analysis using matplotlib shows that Broad Gauge contributes the largest share of the railway network. Metre and Narrow gauges form a much smaller portion in comparison. Overall, it clearly highlights the dominance of Broad Gauge in the system.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

In this Scenario, Generate percentage features for Broad, Metre, and Narrow gauges using pandas and calculate yearly growth in total tracks with NumPy. Use matplotlib to create line and stacked bar charts for trend analysis. This helps understand how railway usage and gauge composition evolve over time.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function <code>np.diff</code> to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Graph :

The line graph using matplotlib displays how Broad, Metre, and Narrow gauges evolve over the years. It makes it easy to observe increases, decreases, or stable trends for each gauge type.

The stacked bar chart created using matplotlib shows the composition of Broad, Metre, and Narrow gauges for each year. Each bar represents total tracks, with different segments showing each gauge's contribution.

Output : Line Graph

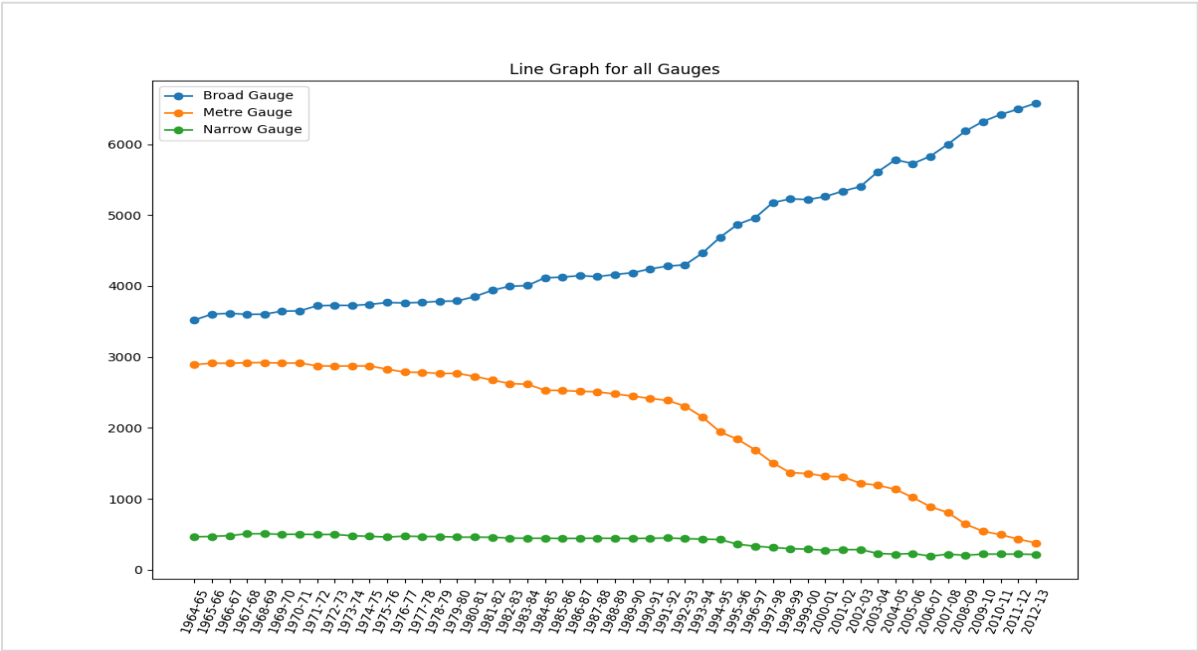


Fig 4: Line Graphs for all Gauges

Output : Stacked Bar Chart

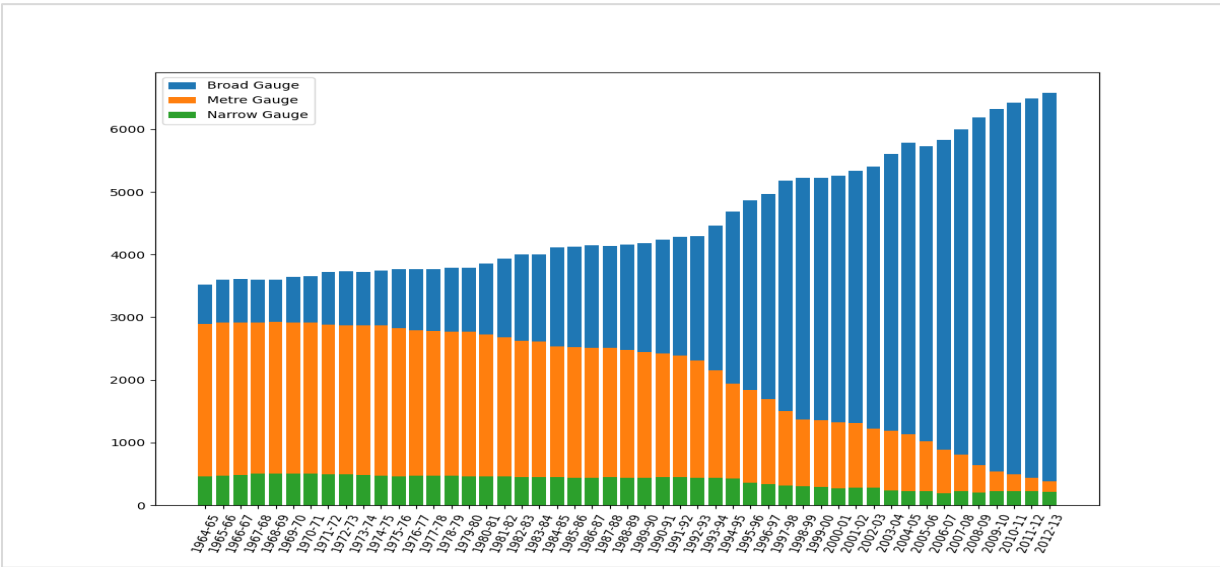


Fig 5: Gauge Composition

Conclusion:

This analysis using matplotlib and pandas highlights a clear long-term shift in railway infrastructure. Broad Gauge consistently dominates across all years and continues to grow steadily. Metre and Narrow gauges show a declining or stagnant trend, indicating reduced usage. The stacked and line charts together confirm the gradual replacement of smaller gauges. Overall, the system is moving toward standardization with Broad Gauge as the primary network type.

Final Conclusion:

This project using pandas, NumPy, and matplotlib provides a complete analysis of railway gauge trends from data cleaning to advanced visualization. The results consistently show that Broad Gauge dominates across all years and continues to grow steadily. Metre and Narrow gauges gradually decline in both usage and contribution. Growth analysis and visual charts clearly highlight a shift in railway structure over time.

What you have learned

I learned how to load, clean, and transform real-world datasets effectively. I practiced handling missing values, converting data types, and creating new features for deeper analysis. I also gained experience in visualizing trends using line graphs, bar charts, and pie charts to extract meaningful insights. Finally, I learned how to interpret data patterns and present clear conclusions based on visual evidence.